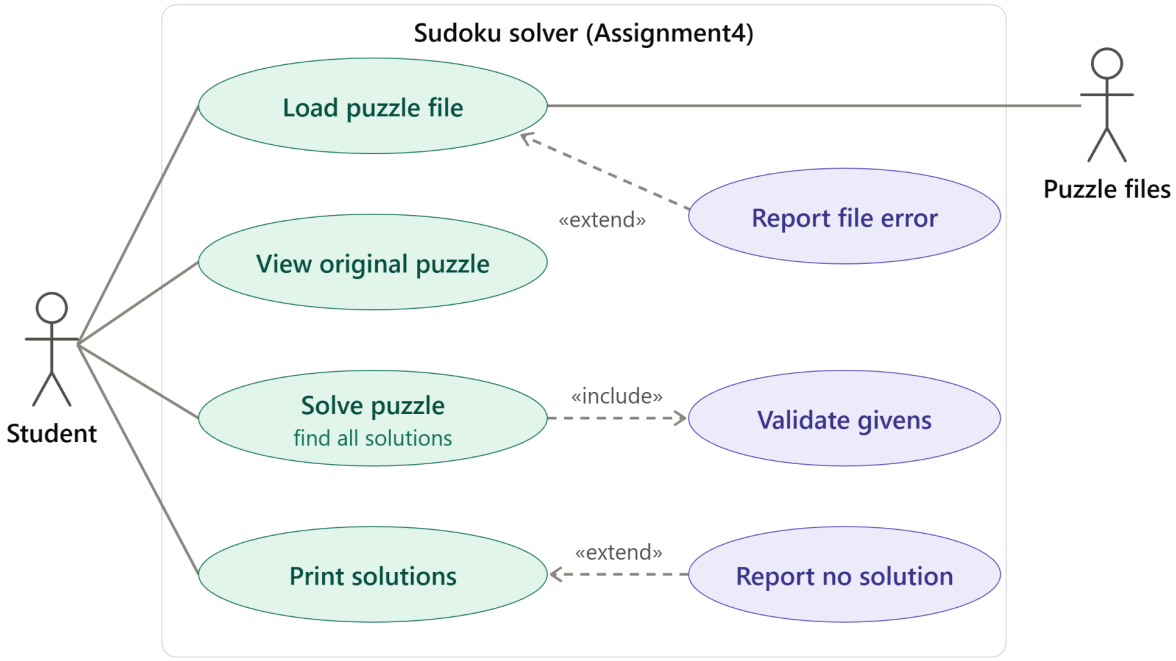


Use Case: Solve Puzzle

The student runs the Sudoku Solver, which reads the puzzle files, checks the numbers, and finds all possible solutions using backtracking.

Step by step
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1. Student runs program
2. System loads the puzzle file
3. System displays original puzzle
4. System checks if the starting numbers are valid
5. System uses recursive backtracking to solve puzzle
6. System prints all solutions found
7. System repeats for the remaining puzzles

Detailed Description: The program's preconditions are making sure the program is compiled and the sudoku puzzle files are available. The main flow is the program reads each sudoku file, checks the starting numbers, and uses backtracking to find all valid solutions. Alternative routes would be (file connote be read, display error message), (if no solution exists, system prints “no solution found”) and (if multiple solutions exist, the system prints all of them)